Business Analytics

ENGLISH

Specialization BSc Wirtschaftsinformatik / Business Information Technology

Designed for:

students interested in **advanced business analytics** in business information technology.

Programme focus:
- Machine Learning
- Big Data Analysis, Analysis of Semi- and Unstructured Data
- Time Series Analysis
- Analytics Project

Recommended prerequisites:

Statistics and Probability

Business Intelligence

Machine learning with Python *

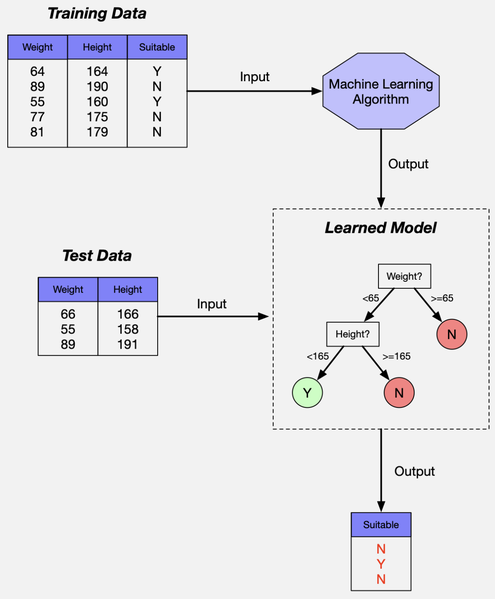
Machine learning is about automatically learning knowledge from data.

Module Content:

- Introduction to Python:
 Basics, Data Wrangling, Programming, Communication
- Machine Learning:
 Basics, Supervised Learning, Unsupervised Learning, Association Analysis

Application examples:

- Automatic recommendations on which movies to watch
- Personal verification using fingerprints
- Recognition of friends in photos
- Fraud detection



Time Series Analysis with R *

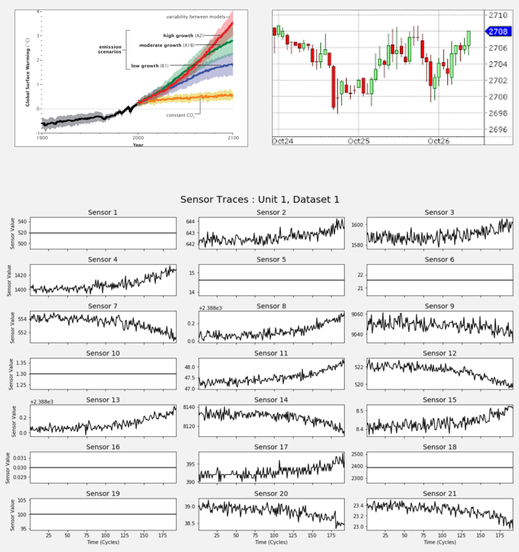
Time Series describe the change of data over time.
Time Series Analysis touches every other area of machine learning.

Module Content:

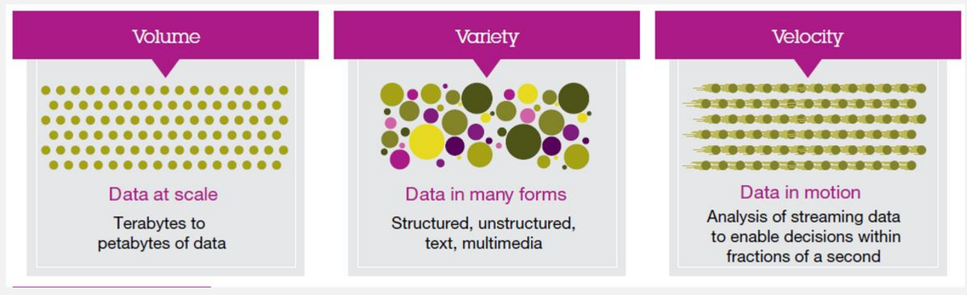
- Introduction to R:
 Basics, Data Wrangling, Programming, Communication
- Time Series Analysis:
 Component models, Regression models, Exponential smoothing models, ARIMA models

Application examples:

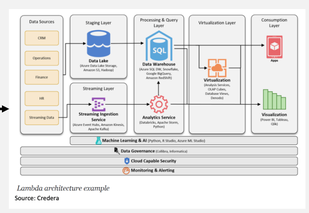
- Smart Health Care Products
- Predictive Maintenance
- Climate Change Modelling
- Stock Market Analysis and Algorithmic Trading



Big, Semi- and Unstructured Data *



- Volume: Distributed Storage / MapReduce
- (semi-structured) text data:
 - Information Extraction, Named Entity Recognition
 - Information Retrieval / Semantic Search
 - Text Mining
 - Sentiment Analysis
- Velocity: Stream Processing, Complex Event Processing
- Architectures: Data Lakes, Lambda/Kappa architectures
- Cloud platforms und workflows



Lambda architecture example
Source: Credera

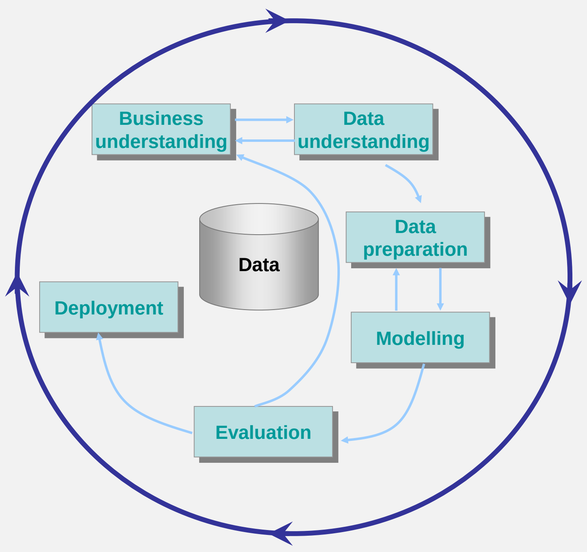
Analytics Project

Students gain practical experience in a larger analytical project.

The goal is to pass all the phases of an analytics project.

As part of a team, students
- begin with the business problem understanding,
- With data collection, data preparation and data cleaning,
- find the right machine learning algorithm,
- evaluate it and present the findings in an understandable way according to CRISP DM methodology.

CRISP-DM Reference model



After graduation:

Data Analyst	You collect and analyze the available data, show correlations and create reports for strategic planning.
Data Scientist	You ensure a solid database, develop new algorithms to find correlations and create forecasting models.
Data Engineer	You are involved in setting up data warehouses, ensuring data flows and advising specialist departments on data handling.

* Can also be chosen as an elective module

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